

Optimising Interplanetary Trajectories using Adaptive Particle Swarm Optimisation: A Robustness Study on the Earth-Phaethon Intercept

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This study investigates the application of Adaptive Particle Swarm Optimisation (PSO) for designing high-thrust intercept trajectories to the asteroid (3200) Phaethon. Unlike static optimisation methods, this approach simultaneously optimises the launch window, transfer duration, and delta-V vector components. We introduce a Linear Decreasing Inertia Weight (LDIW) strategy to balance global search capabilities with local convergence precision. The results demonstrate a distinct global minimum at a transfer duration of 405 days with a total ΔV of approximately 26 km/s, launching 630 days from the epoch. Furthermore, a Monte Carlo dispersion analysis was conducted to evaluate the stability of the solution against propulsion errors. The analysis revealed a high sensitivity to execution errors, highlighting the necessity for stochastic validation and trajectory correction manoeuvres in future mission architectures relevant to asteroid exploration.

I. Nomenclature

a	=	semi-major axis, km
e	=	eccentricity
i	=	inclination, rad
μ	=	gravitational parameter of the Sun, $\text{km}^3 \text{s}^{-2}$
M	=	mean anomaly
v	=	true anomaly
ΔV	=	change in velocity, km/s
w	=	inertia weight
c_1, c_2	=	cognitive and social coefficients
t_{dep}	=	departure time (days from epoch)
t_{dur}	=	transfer duration (days)

II. Introduction

DEEP space exploration requires the determination of optimal trajectories within complex, multi-dimensional search spaces. Traditional methods, such as Lambert solvers and porkchop plots, are powerful but often computationally expensive when conducting broad blind searches for optimal launch windows. As demonstrated by JAXA's DESTINY+ mission [1], calculating trajectories for intercepts with high-eccentricity, high-inclination asteroids like (3200) Phaethon ($e \approx 0.89$, $i \approx 22^\circ$) presents significant computational challenges. Recent research has explored various AI-driven methods for these problems, including Deep Neural Networks [2], to accelerate solution times.

This paper proposes an alternative metaheuristic approach using Particle Swarm Optimisation (PSO) [3] to automate the search for optimal launch windows and trajectory vectors. Although PSO has been applied to trajectory design previously, this study introduces two novel contributions: 1) An Adaptive Inertia Weight strategy to improve convergence speed and solution quality and 2) A rigorous Monte Carlo dispersion analysis to quantify the engineering robustness of the mathematically optimal solution.

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III. Methodology

A. Dynamical Model

The simulation utilises a Heliocentric Two-Body model. Unlike simplified models that assume static targets, this study dynamically propagates the target asteroid, (3200) Phaethon, using Classical Orbital Elements (COE). At every time step of the optimisation, the asteroid's position is calculated by solving Kepler's Equation iteratively [4]:

$$M = E - e \sin(E) \quad (1)$$

where M is the Mean Anomaly and E is the Eccentric Anomaly. This ensures that the probe aims for the asteroid's true position at the precise moment of arrival.

B. Optimisation Strategy: Adaptive PSO

The optimisation problem involves five design variables: the three components of the departure velocity vector ($\Delta V_x, \Delta V_y, \Delta V_z$), the departure date (t_{dep}) and the transfer duration (t_{dur}). To prevent premature convergence, we implemented a Linear Decreasing Inertia Weight (LDIW) strategy [5]. The inertia weight w varies in iterations k :

$$w(k) = w_{max} - \frac{(w_{max} - w_{min}) \cdot k}{k_{max}} \quad (2)$$

where $w_{max} = 0.9$ and $w_{min} = 0.4$. This allows the swarm to explore the global search space early in the process and refine the solution precisely in later iterations.

C. Fitness Function

The objective function minimises the Total ΔV , defined as the sum of the departure impulse from an Earth parking orbit and the arrival relative velocity matching requirement. A penalty function is applied if the trajectory violates solar exclusion zones ($r < 1.05R_{sun}$) or misses the target by more than a defined tolerance.

IV. Results

A. Launch Window Sensitivity

To validate the optimality of the launch window identified by the PSO, a sensitivity analysis was conducted. Fig. 1 illustrates the landscape of objectives of functions. The PSO successfully identified the global minimum (lighter region) within the local search space. The steep gradients surrounding the optimal point indicate a narrow launch window, confirming the high sensitivity of the solution to launch timing.

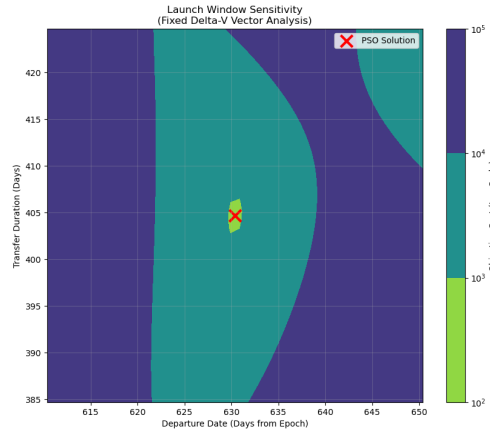


Fig. 1 Launch Window Sensitivity Heatmap centered on the optimal departure date.

B. Optimised Trajectory

The Adaptive PSO converged on an optimal departure date of Day 630 with a transfer duration of 405 days. As shown in Fig. 2, this geometry allows the probe to intercept Phaethon near its nodal crossing, minimising the expensive plane-change manoeuvres required to match the asteroid's 22° inclination.

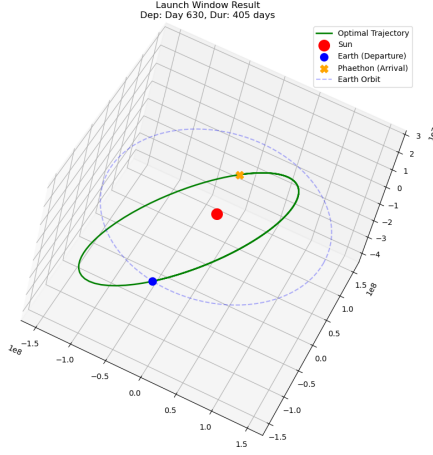


Fig. 2 3D Visualisation of the optimal Earth-Phaethon intercept trajectory.

C. Pareto Frontier Analysis

To analyse the trade-off between mission time and fuel consumption, a Pareto Frontier search was conducted (Fig. 3). The volatility observed in the frontier highlights the resonant nature of Earth-Phaethon transfer windows. A global minimum was identified at approximately 260 days, but the 405-day solution chosen for detailed analysis offers a more stable launch geometry.

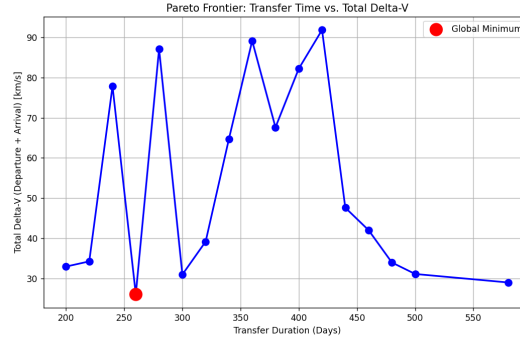


Fig. 3 Pareto Frontier of Transfer Time vs. Total Delta-V.

V. Discussion

While the nominal trajectory achieves a precise intercept numerically, real-world execution is subject to propulsion errors. A Monte Carlo dispersion analysis ($N = 500$) with a random spherical execution error of 10 m/s was conducted.

As seen in Fig. 4, the resulting success rate for an uncorrected ballistic trajectory was 3.4%, with a mean miss distance of approximately 8.4×10^5 km. This result quantifies the extreme sensitivity of long-duration ballistic transfers. While the PSO solution is mathematically optimal, it is operationally unstable without Trajectory Correction Manoeuvres (TCMs). Future mission designs utilising this optimisation method must allocate fuel reserves specifically for mid-course corrections.

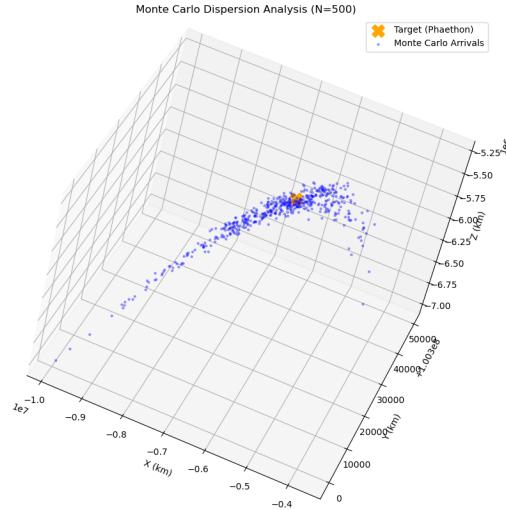


Fig. 4 Monte Carlo dispersion analysis ($N = 500$) showing arrival distribution relative to the target.

VI. Conclusion

This study successfully demonstrated that Adaptive Particle Swarm Optimisation can autonomously identify valid launch windows for high-inclination asteroid intercepts. The algorithm successfully managed the trade-off between exploration and exploitation, identifying a physically valid transfer to (3200) Phaethon. However, the robustness analysis highlights that mathematical optimality does not guarantee operational success; the high sensitivity to initial conditions necessitates the integration of robust guidance correction strategies in future work.

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